

Advanced Programming Project, Software Engineering 3rd Year, Section A and B,
Project Title: Multi-user Chat Application, Weight: 10%, Deadline: 11/05/2026

Project Description: A client-server based real-time chat system that allows multiple users to send/receive text messages and share files. Your project will have the following modules: User authentication, File Transfer Module, and Multithreading Module

1. **User authentication:** provides user registration/signup Javafx interface that allows users to register their user_id, full_name, and password. The system should store this information on a **ChatDatabase**. In addition to the sign up interface the module should also have a login window that allows registered users to login into the system.
2. **File Transfer Module:** allow users to select and transfer files via JavaFX FileChooser to the server. Transferred files (with attribute sender, receiver (if private), file name, date transferred) should be stored in **ChatDataBase**.
3. **Multi-threading Module:**
 - 3.1. **Server:** should accept each client on a new thread. It should use separate threads for handling file transfers to avoid UI freeze.

Tasks of the server

- Accept new client connections
- Handle each client independently (concurrent communication)
- Receive messages from one client and broadcast/forward to others
- Handle file transfers
- Maintain a list/map of active clients
- Close connections when clients disconnect

Server Threads:

a) Main Server Thread:

- Listens for new connections using ServerSocket.accept()
- Assigns a new thread for each incoming client.

b) Client Handler Threads (One per Client):

- Continuously listen for messages from a specific client.
 - Process and relay messages to the appropriate target (group/private).
 - Handle file transfer.
- 3.2. **Client:** Should have a thread for listening to incoming messages and updates. It should use background thread for file sending/receiving.

Tasks of Client:

- Connect to the server via a socket
- Send messages typed by the user to the server
- Receive and display incoming messages from the server

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- Handle file transfer
- Maintain responsive GUI (JavaFX)

Client Threads:

a) Main JavaFX UI Thread:

- Renders the GUI and handles user interactions (list of active users, TextArea for input, message panel, buttons etc)
- Active Users: Should have a panel that shows a list of active users (who are currently logged in to the system).
- Chat Display Panel: a scrollable panel that displays messages from server
- Chat Writing Box: a text area that allows a user to type in messages

b) Message Listener Thread:

- Continuously reads messages sent from the server (readUTF())
- Updates the GUI from background using Platform.runLater()

c) File Transfer Thread: Handles sending or receiving files without freezing the GUI

Note: Create a MySQL database named **ChatDatabase** to store messages and transferred files.

The database should have at least the following tables:

- **users:** user_id, full_name, password, email
- **messages:** id, sender_id, receiver_id, message, timestamp
- **files:** id, sender_id, receiver_id, file_name, file_size, timestamp